LED BOX MODIFICATION

User Electrical Information Notes

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i) Referenced Documents

[1] Raspberry Pi Pico Datasheet, Raspberry Pi Ltd.

[2] SNx4AHCT125 Quadruple Bus Buffer Gates With 3-State Outputs Datasheet, Texas Instruments.

[3] Rigol DM3058E Digital Multimeter Data Sheet, RIGOL Technologies Inc

ii) Abbreviations Used

DC	:	Direct Current
DPDT	:	Double Pole Double Throw
GND	:	Ground (O V reference)
GPIO	:	General Purpose Input/Output
IC	:	Integrated Circuit
I/F	:	Interface
I/O	:	Input/Output
LED	:	Light Emitting Diode
LiPo	:	Lithium Polymer
PSU	:	Power Supply Unit
STR	:	Systems Theory and Robotics Group
USB	:	Universal Serial Bus
UV	:	Ultraviolet
VCC	:	Voltage at the Common Collector (Positive Supply Voltage)
VOH	:	Voltage Output High

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Table B1: Component Type Summary

1. Introduction

This document gives an overview of the circuitry modifications made to the LED Box unit to meet requested upgrade requirements.

The LED Box unit is utilised by the Systems Theory and Robotics (STR) group as part of experimental work.

2. Warnings

2.1 Ultraviolet Light

The LED Box unit utilises Light Emitting Diodes (LEDs) that produce radiation within the ultraviolet (UV) spectrum. These LEDs are the same devices used in the original LED Box design, as it was a requirement that they remain unchanged.

No technical documentation or records identifying the specific LED types are available. Consequently, no manufacturer datasheets can be referenced to determine the exact emission wavelengths or radiant output levels.

Prior to use of the LED Box in any experimental setup, the level of ultraviolet radiation emitted must be measured. A formal Risk Assessment should then be conducted based on the measured output to ensure that any potential hazards are identified and that appropriate control measures are implemented.



2.2 Lithium Polymer Battery

The LED Box unit is powered by a Turnigy 9XR 3-cell (3S) 11.1 V Lithium Polymer (LiPo) battery pack with a capacity of 2200 mAh. This is the same battery specification used in the original LED Box, as the power source arrangement was required to remain unchanged.

The battery incorporates built-in overcurrent and overcharge/over-discharge protection. For charging, the battery is removed from the LED Box enclosure and charged using its specified dedicated charger.

Prior to using the LED Box in an experimental setup, a Risk Assessment should be performed to ensure that any hazards that may compromise the LiPo battery safety are adequately addressed.



3. LED Box Modification Design Requirements

The following design modification requirements for the existing LED Box were requested and agreed:

- Retain existing enclosure, LED layout and the existing 24 LEDs.

- LEDs to be set up to the same drive current as in existing LED Box.
    - ⇨ Existing LED Box has LED currents between 7mA to 8mA. Therefore, LEDs to be set to nominal 7.5mA each. Aim will be to provide some range of LED current adjustability through use of a trimpot, in case the LED current needs change or be adjusted in the future.

- The existing LiPo battery power source arrangement to remain the same. There was discussion of options like external power, removing/changing the LiPo battery arrangement and possible addition of a power indicator but these were not warranted.
    - ⇨ An accessible fuse in line with battery will be added to provide to additional overcurrent protection.

- A Raspberry Pi Pico microcontroller to be incorporated into the LED box to allow independent control of the 24 LEDs.

- The USB port of the Raspberry Pi Pico is to be made available on the side of the LED Box, to allow ease of access for programming.

- Raspberry Pi Pico GPIOs to be used are 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22 and 26. The LEDs are to be wired consecutively in an orderly layout.

- 74AHCT125 buffer integrated circuits to be used to drive the LEDs. The LEDs are to be wired directly to the buffer output with current limiting resistance to be wired between LED cathode and ground.

4. Original LED Box Details

Figures 1 and 2 show photos of the existing LED Box external features. While Figures 3 and 4 show photos of the existing LED Box internal construction and wiring.

Tracing through the circuit wiring it was found that the LEDs are wired as three strings of eight LEDs. Each LED string is fed through a 100Ω resistor from a DC-DC converter which has been set to a voltage of 25V.

The DC-DC converter is powered by a Turnigy 9XR 3-cell (3S) 11.1 V Lithium Polymer battery pack through a switch (on/off switch). Figure 5 shows the wiring diagram traced out for the existing LED Box.

The measured currents for the LED strings ranged between 7mA and 8mA (two strings at 8mA and one at 7mA).

Based on these measurements, and to meet the requirement of maintaining the same LED current in both the existing and modified LED Box designs, the current will be set to 7.5 mA.

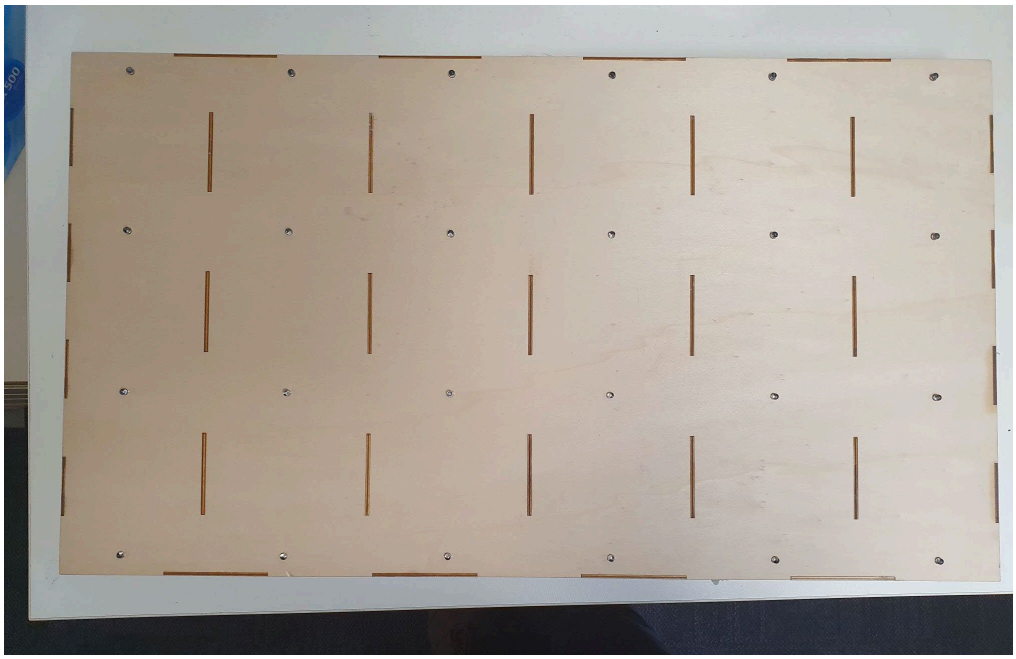


Figure 1: LED Box Top View Before Modification



Figure 2: LED Box Front View Before Modification

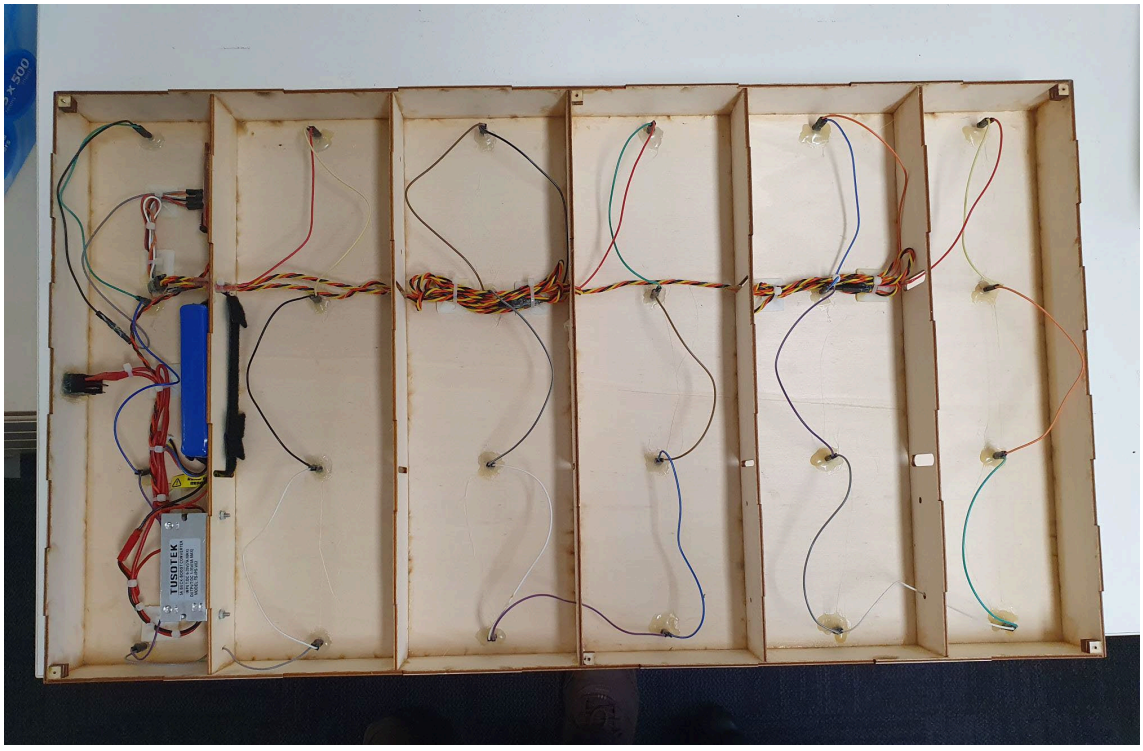


Figure 3: LED Box Internal View Before Modification

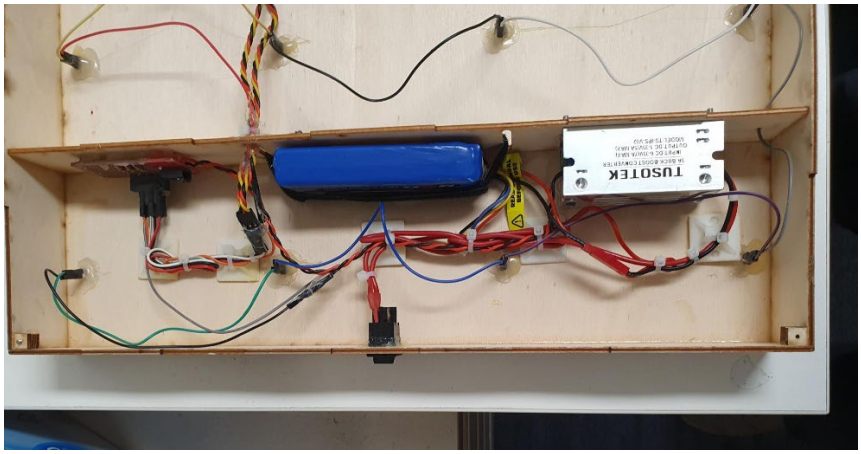


Figure 4: LED Box Internal View Before Modification (Power Section Closeup)

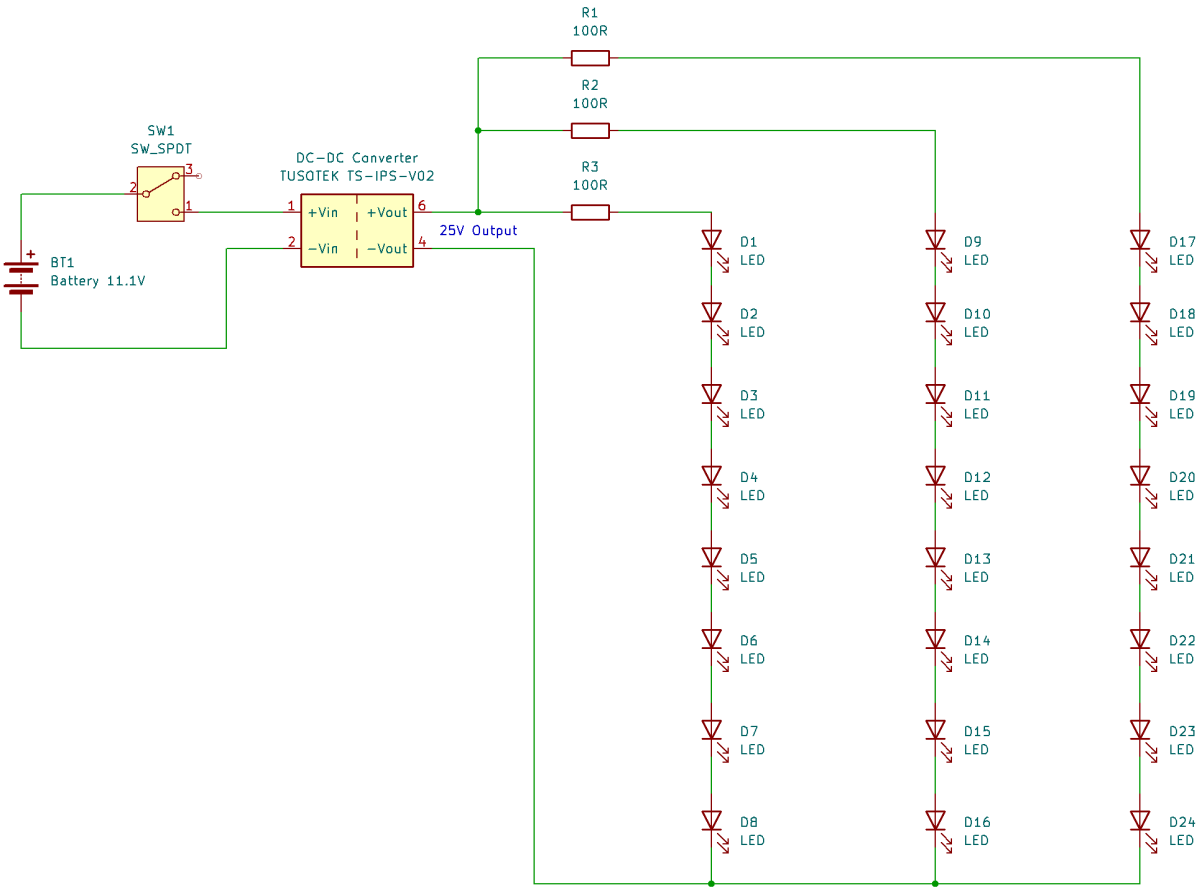


Figure 5: LED Box Circuit Wiring Before Modification

5. Modified LED Box Details

To implement the Modified LED Box three types of boards were designed and constructed: a Power Distribution Board, a Microcontroller Board and an LED Driver Board. These boards are described in the sections that follow.

Section 5.1 presents the Modified LED Box block diagram showing the general interconnections between Power Distribution, Microcontroller and LED Driver Boards. The complete circuit schematic can be found in Section 6 (Appendix A).

Figures 6 to 8 show the top, front and side views of the Modified LED Box, respectively. Figure 9 shows the internal view of the LED Box.



Figure 6: LED Box Top View After Modification



Figure 7: LED Box Front View After Modification

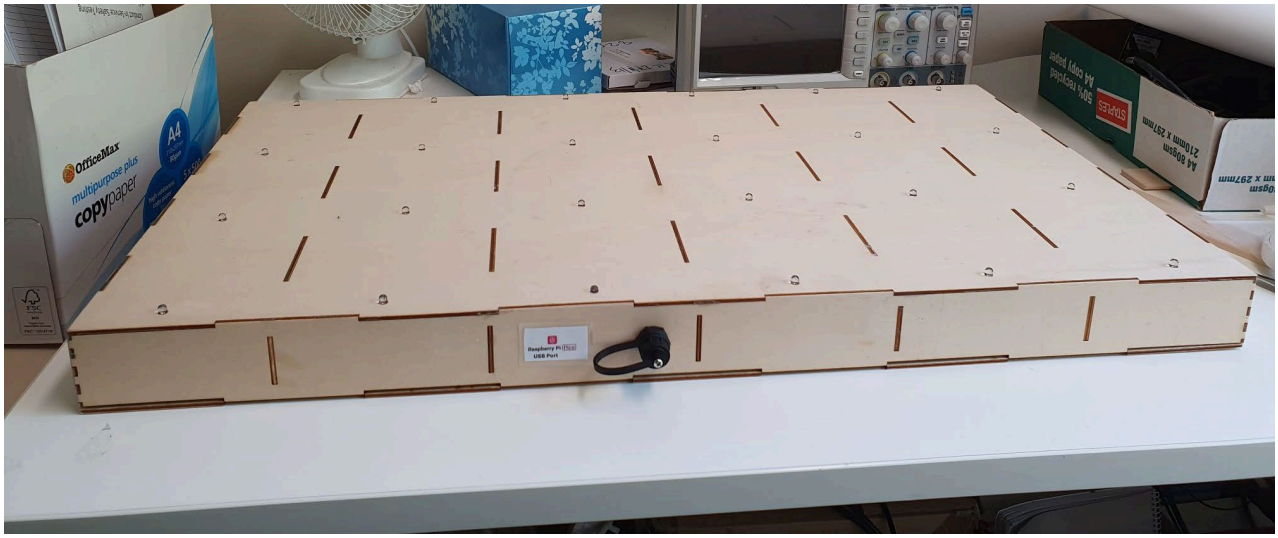


Figure 8: LED Box Side View After Modification

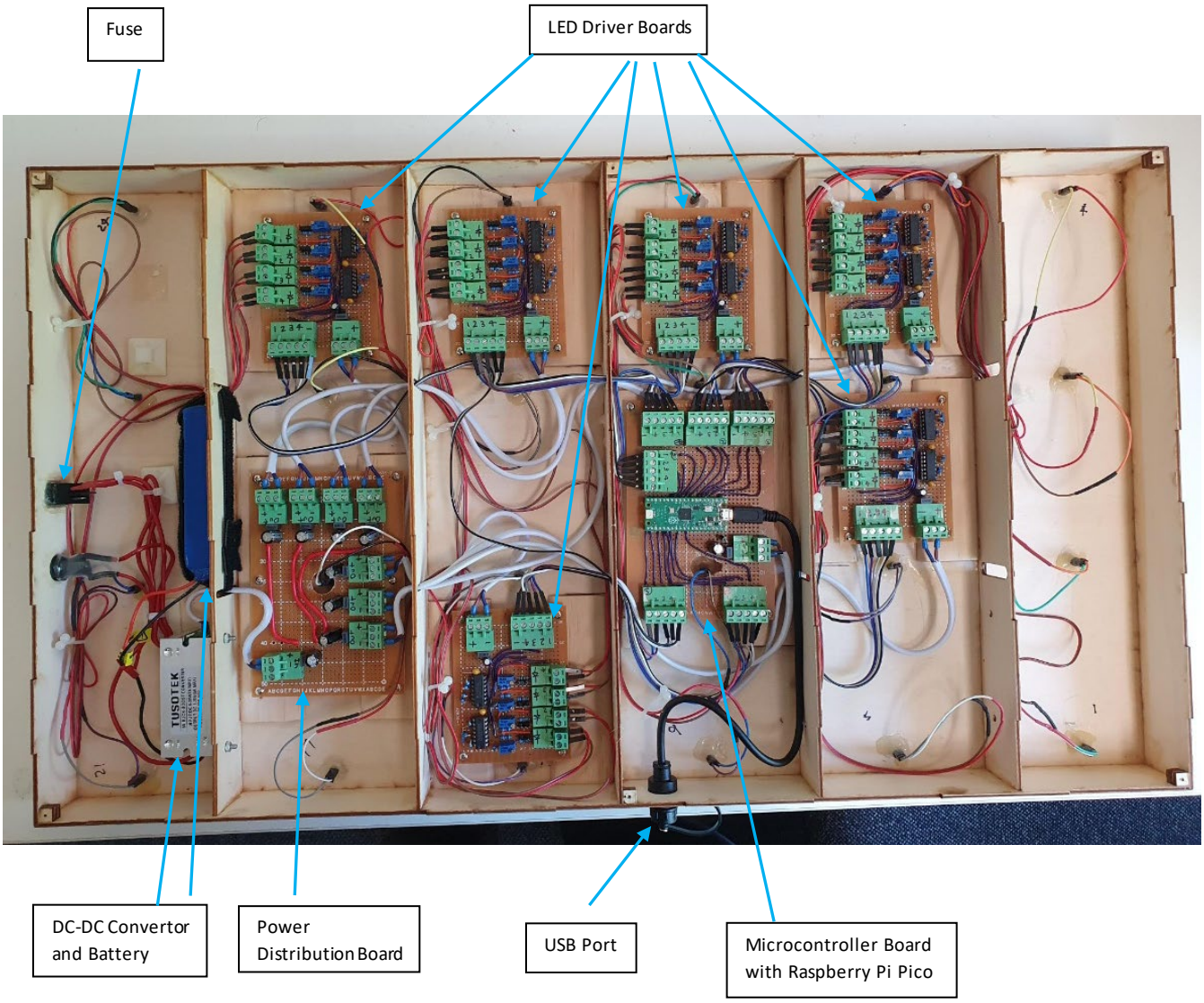


Figure 9: LED Box Internal View After Modification

5.1 Block Diagram

The Modified LED Box block diagram is shown in Figure 10. The Power Supply Block distributes power to seven boards: the Microcontroller Board and six LED Driver Boards.

There are 24 LEDs in total, with each of the six LED Driver Boards supplying current to four LEDs. The Microcontroller Board provides 24 individual control outputs (four for each LED Driver Board). This configuration allows independent on/off control of each LED while distributing the current-driving load across the six driver boards.

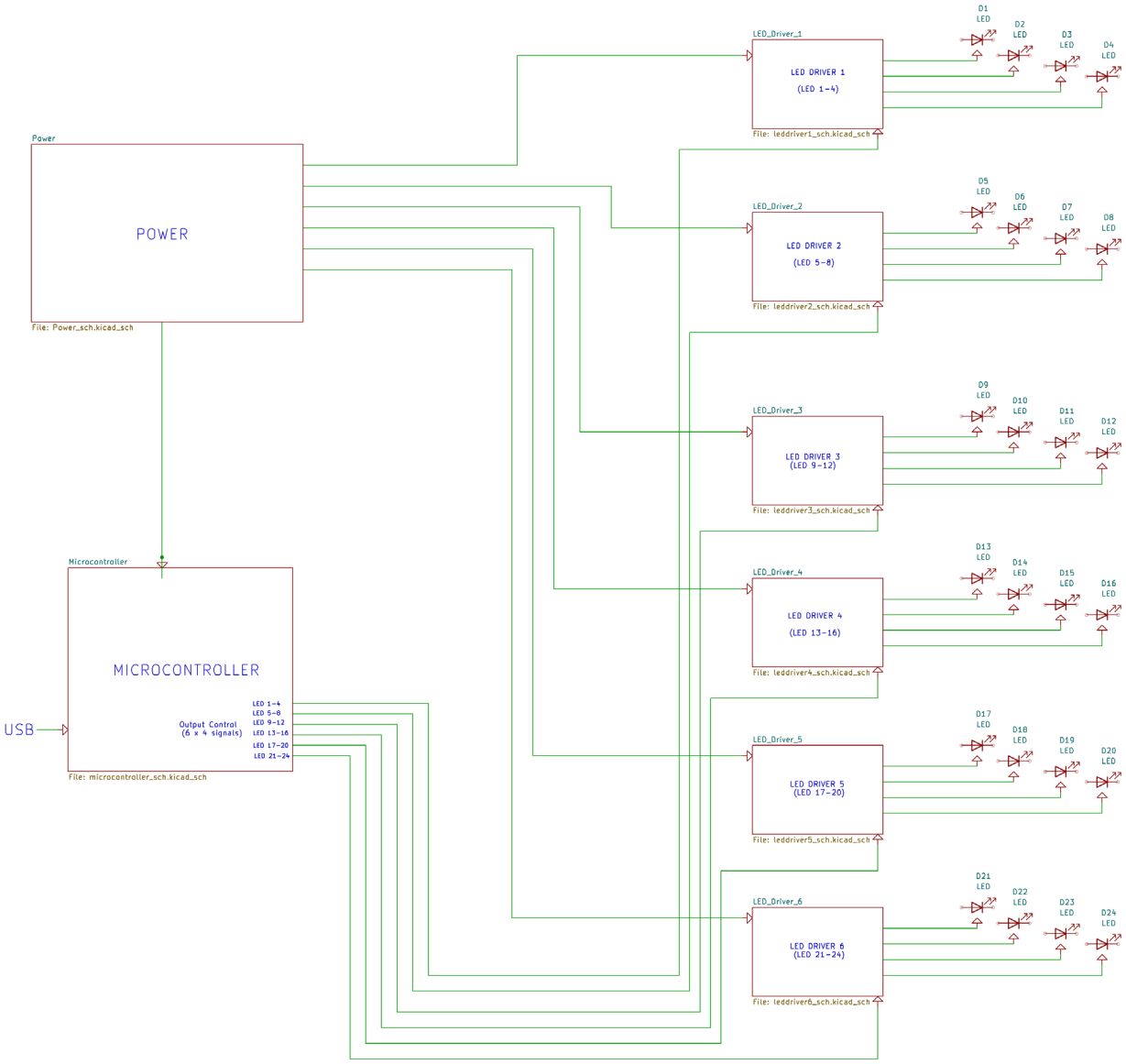


Figure 10: Modified LED Box Block Diagram

5.2 Power Supply

The power supply for the Modified LED Box comprises the existing battery, on/off switch and DC-DC converter arrangement with the addition of an inline protection fuse for the battery and Power Distribution Board shown in Figure 11.

The Power Distribution Board takes the output of the DC-DC converter and provides seven filtered output rails to supply power to all boards within the Modified LED Box.

The output voltage rail of the DC-DC converter has been adjusted to 5.35 V. This compensates for the forward voltage drop caused by the Schottky diode power ORing configuration on the Microcontroller Board (see Section 5.3.1) and the reduced high level output voltage of the 74AHCT125 buffer IC (Integrated Circuit) under load on the LED Driver Boards (see Section 5.4).

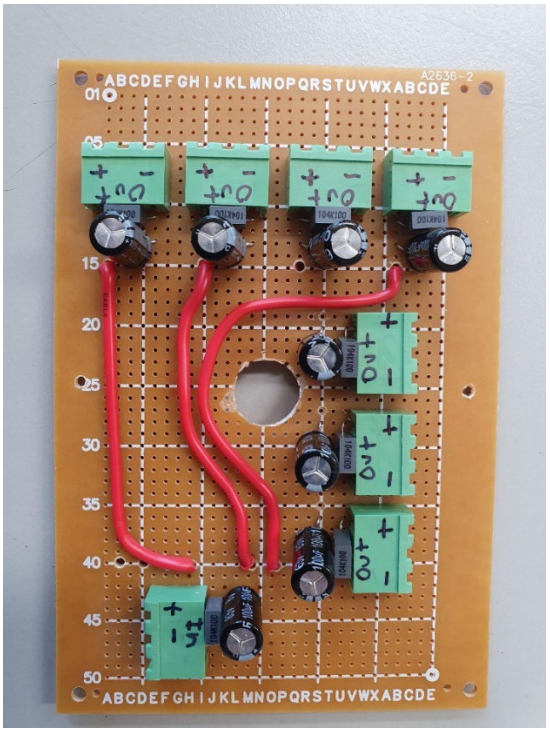


Figure 11: Power Distribution Board

5.2.1 Input Fuse

The input power fuse has been selected to accommodate the full range of adjustable LED operating current, with additional margin to account for the DC-DC converter inrush current.

The fuse is a M205 size T1Amp (Time Delay) fuse.

Figure 12 shows a closeup of the input power fuse holder. To access the fuse, the outer body is rotated counterclockwise and removed as shown in Figure 13. To reinstall the fuse, push it into place and rotate the outer body clockwise until secure.

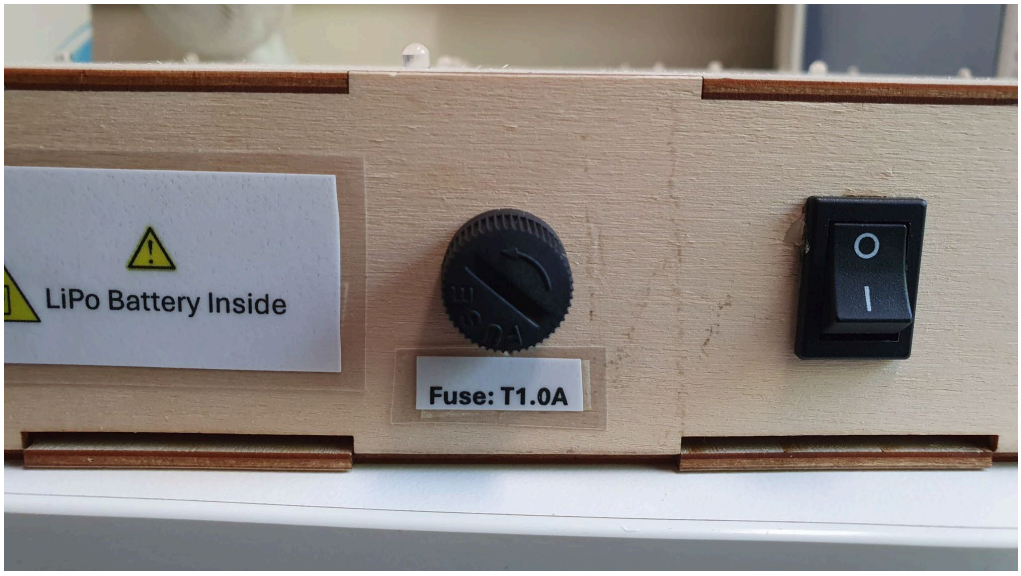


Figure 12: Modified LED Box Input Power Fuse Holder



Figure 13: Modified LED Box Input Power Fuse Holder Open

5.3 Microcontroller

The Microcontroller Board is shown in Figure 14. The Microcontroller Board is built around the Raspberry Pi Pico unit. The Microcontroller Board has:

- Connectors for interfacing to 24 GPIO signals used for LED control
- Power supply filtering components
- A Schottky diode connected to the Pico's VSYS pin.

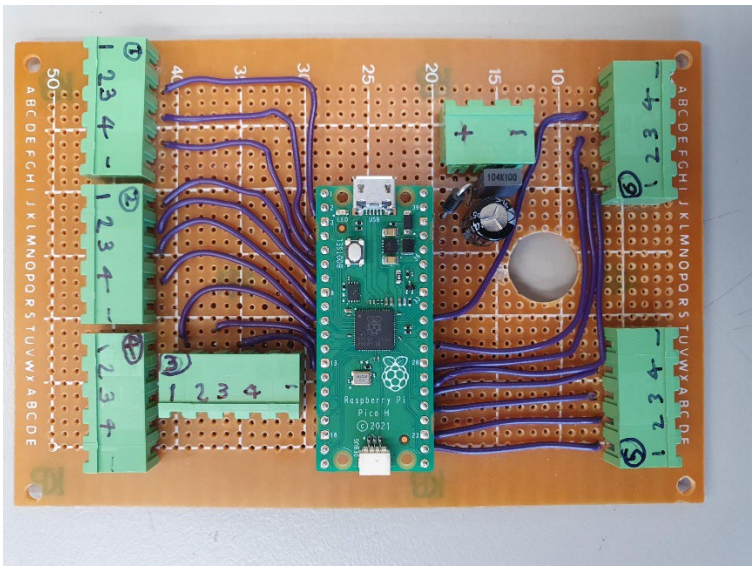


Figure 14: Microcontroller Board

5.3.1 Raspberry Pi Pico Power Supply Arrangement

Powering the Raspberry Pi Pico through the Schottky diode implements a simple power ORing scheme. This allows the Pico to be powered from either:

1. The internal LED Box supply (battery → DC-DC converter), or
2. USB VBUS when connected to a computer.

This arrangement effectively creates an automatic supply selection system where:

- The higher voltage source powers the Pico, and
- The two supplies remain electrically isolated from each other, preventing backfeeding, ensuring that one supply does not drive current into the other.

This provides seamless switching between USB power and the internal battery supply without requiring manual intervention or additional control circuitry. The circuitry can be seen on Page 3 of the Circuit Diagram in Section 6 (Appendix A). Further information on the powering scheme can be found in Section 4.5 of [1], the Raspberry Pi Pico Datasheet.

Note: that the LEDs can only be driven when powered from the internal battery/DC-DC converter setup.

The Raspberry Pi Pico cannot directly supply sufficient current or voltage to power the LED Driver boards and LEDs. Therefore, the LED Drivers and LEDs are powered from the internal battery/DC-DC converter setup.

As a result:

- The Raspberry Pi Pico provides only the control signals.
- The power for the LEDs comes exclusively from the internal battery/DC-DC converter.
- The LEDs will operate only when the internal battery/DC-DC converter system is used.

5.3.2 Raspberry Pi Pico USB Port

Access to the Raspberry Pi Pico USB port is provided via a Micro-B panel-mount USB extension cable. The extension cable can be seen in Figure 9 (Modified LED Box internal view).

Figure 15 shows the Modified LED Box Raspberry Pi Pico USB port with the dust cover installed. Figure 16 shows the same USB port with the dust cover removed to allow USB access.

As described in Section 5.3.1 the Raspberry Pi Pico can be powered through the USB port or internal battery/DC-DC setup for programming activities etc. Although the LEDs can only be driven when powered from the internal battery/DC-DC converter setup.



Figure 15: Modified LED Box Raspberry Pi Pico USB Port (Dust Cover On)



Figure 16: Modified LED Box Raspberry Pi Pico USB Port (Dust Cover Off)

5.3.3 Raspberry Pi Pico GPIO to LED Mapping

- Figure 17 shows the LED numbering layout for the Modified LED Box. The LEDs are numbered sequentially from left to right, starting at the top-left corner. The numbering then continues on the next row below, again proceeding from left to right.

- Table 5.3.3 shows the Raspberry Pi Pico GPIO to LED mapping.

- A logic HIGH will turn a LED on. A logic LOW will turn a LED off.

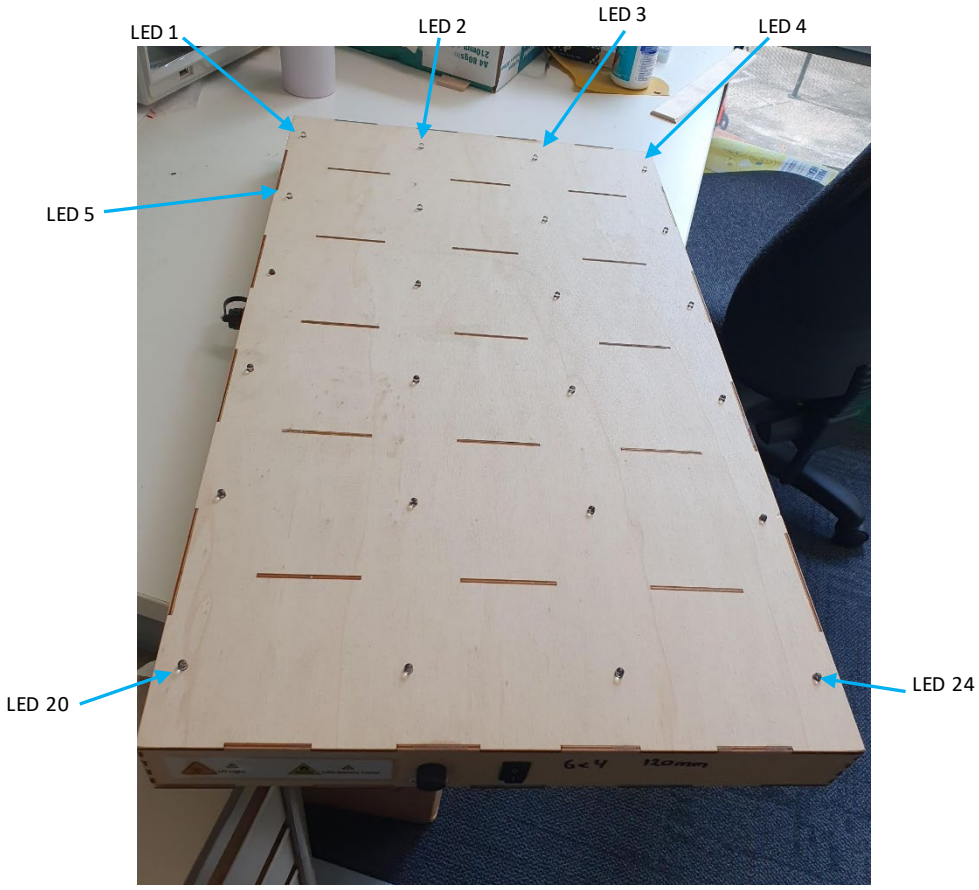


Figure 17: Modified LED Box LED Numbering

LED Number	Raspberry Pi Pico GPIO	Logic
1	GP0	HIGH=LED on, LOW =LED Off
2	GP1	HIGH=LED on, LOW =LED Off
3	GP2	HIGH=LED on, LOW =LED Off
4	GP3	HIGH=LED on, LOW =LED Off
5	GP4	HIGH=LED on, LOW =LED Off
6	GP5	HIGH=LED on, LOW =LED Off
7	GP6	HIGH=LED on, LOW =LED Off
8	GP7	HIGH=LED on, LOW =LED Off
9	GP8	HIGH=LED on, LOW =LED Off
10	GP9	HIGH=LED on, LOW =LED Off
11	GP10	HIGH=LED on, LOW =LED Off
12	GP11	HIGH=LED on, LOW =LED Off
13	GP12	HIGH=LED on, LOW =LED Off
14	GP13	HIGH=LED on, LOW =LED Off
15	GP14	HIGH=LED on, LOW =LED Off
16	GP15	HIGH=LED on, LOW =LED Off
17	GP16	HIGH=LED on, LOW =LED Off
18	GP17	HIGH=LED on, LOW =LED Off
19	GP18	HIGH=LED on, LOW =LED Off
20	GP19	HIGH=LED on, LOW =LED Off
21	GP20	HIGH=LED on, LOW =LED Off
22	GP21	HIGH=LED on, LOW =LED Off
23	GP22	HIGH=LED on, LOW =LED Off
24	GP26	HIGH=LED on, LOW =LED Off

Table 5.3.3: Raspberry Pi Pico GPIO Mapping

5.4 LED Driver

The LED Driver Board is shown in Figure 18. Each board is capable of independently driving four LEDs. Therefore, as seen in Figure 9 a total of six boards are required to drive the 24 LEDs in the LED Box. The complete circuit diagram is provided in Section 6 (Appendix A). Key design points are:

- ⇨ The LED Driver Board utilises the 74AHCT125 buffer IC, as specified in the design requirements. According to the manufacturer's specifications [2], the device has a recommended maximum output current of 8 mA per output, with an absolute maximum of 25 mA per output and 50 mA per chip. To ensure operation within these limits, two 74AHCT125 ICs are used per board. However, only two of the four available buffers in each IC are utilised, thereby keeping the total output current per chip within the allowable range.

- ⇨ The design requirements specify that the LED current limiting resistor must be placed between the LED and GND. This configuration is shown in the circuit diagram (Section 6, Appendix A) and in Figure 19, which provides an enlarged view of a single LED drive output circuit.

  As shown, the current limiting network consists of a 100Ω fixed resistor in series with a 1kΩ, 25 turn trimpot. This series arrangement allows the LED current to be adjusted over a defined range while ensuring a minimum resistance is always present to prevent excessive current.

- ⇨ As discussed in Section 5.2, the LED driver boards are powered from a 5.35 V supply. Although slightly above the nominal 5 V level, this remains within the 5.5 V maximum recommended supply voltage. Operating at 5.35 V improves the high level output voltage of the 74AHCT125 buffer IC, which tends to decrease significantly under load conditions.

  The 74AHCT125 datasheet [2] specifies that with an output load current of 8 mA, the output high voltage (V_{OH}) may be up to 0.7 V below the applied supply voltage V_{CC}.

- ⇨ 10kΩ pulldown resistors have been used on all buffer inputs. This ensures that buffer inputs do not float during high-impedance states. The pull-downs also improve noise immunity, which is particularly important in the Modified LED Box where many control signal connections span relatively long distances – due to the size of the box. The enable pins have 10kΩ pulldown or pullup resistors as required.

⇨ The physical LED Driver board has been implemented using IC sockets for all buffer devices. This allows devices to be easily replaced if required.
In addition, all pullup and pulldown resistor to buffer IC connection nodes have been made accessible on the resistor lead. This allows:

- A clip point for signal monitoring
- Ability to force of signals HIGH or LOW for debugging
- "Cut-and-strap" modifications

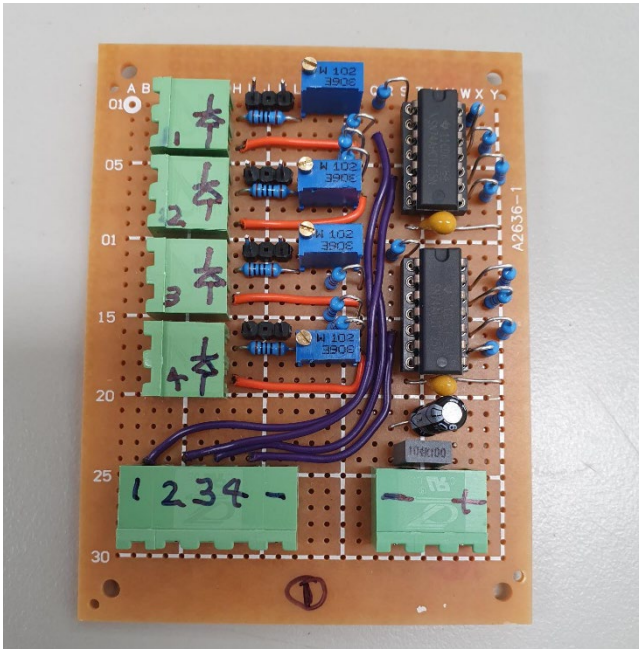


Figure 18: LED Driver Board

5.4.1 LED Current

5.4.1.1 LED Current Adjustment

Figure 19 shows the configuration of the LED drive output circuit. The LED current limiting network consists of a 100Ω ±1% fixed resistor in series with a 1kΩ, 25 turn trimpot.

The 1kΩ, 25 turn trimpot allows LED current adjustment. The 100Ω resistor has a 2 pin header connector across it. This allows measurement of the voltage drop across the 100Ω resistor and hence a determination of the LED current using Ohm's law. i.e LED current = $V_{100\Omega\ resistor}$/100.

The 100Ω resistor header connectors and adjustment trimpots are shown in Figure 20.

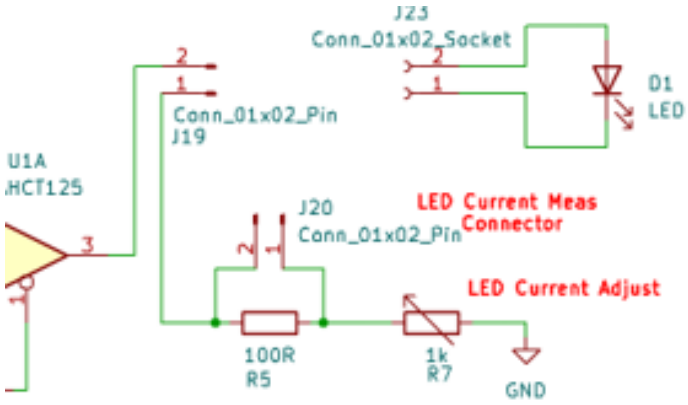


Figure 19: LED Drive Output Circuit

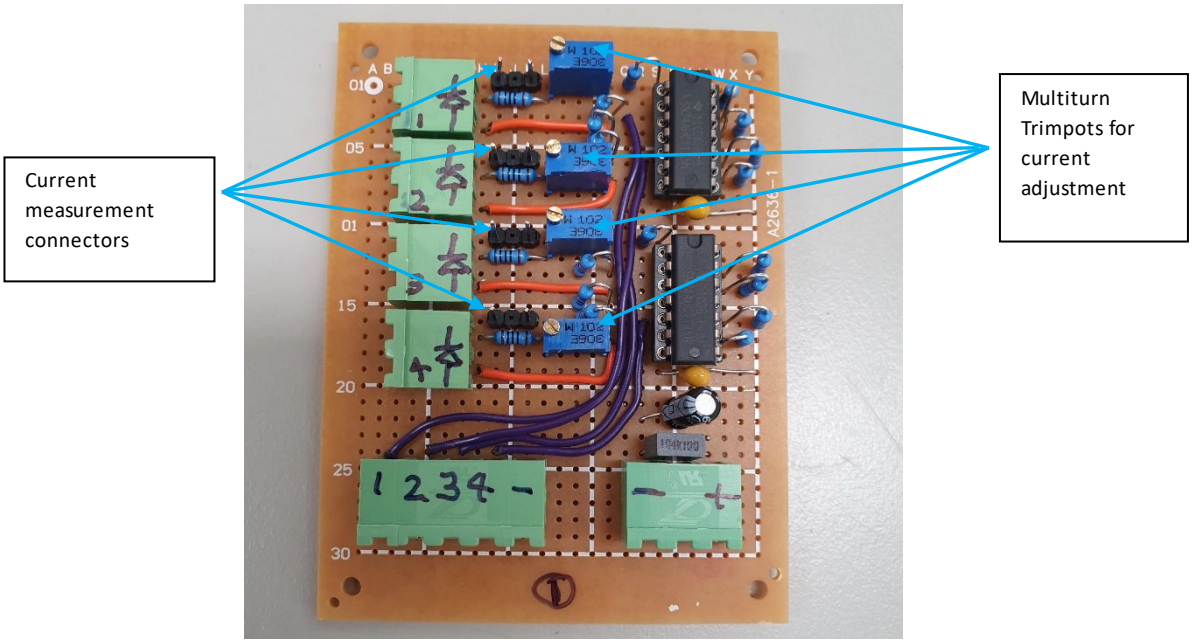


Figure 20: LED Driver Board Showing Current Measurement and Adjustment Points

To meet the design requirement of the LEDs being driven at the same current as existing LED Box of (7 - 8 mA), all the LEDs have been set to a nominal 7.5mA current i.e 0.750 V when measured across the 100Ω ±1% fixed resistor.

A Rigol DM3058E 5 ½ digit bench multimeter was used to measure the 100Ω resistor voltage drop. The DM3058E has an accuracy of ± (0.015% Rdg + 0.003%) on the 2.00000 V Range used. With this accuracy specification the DM3058E will introduce very little error in the measurement. The DM3058E contribution to total uncertainty is negligible compared to the resistor tolerance of ±1%. See calculation in Appendix C.

In practice, also considering the adjustment granularity of the 25 turn trimpot, the achievable current setting is approximately: **7.5± 0.1 mA**. This comfortably satisfies the design requirement of operating within the 7–8 mA range.

5.4.1.2 LED Current Adjustment Range

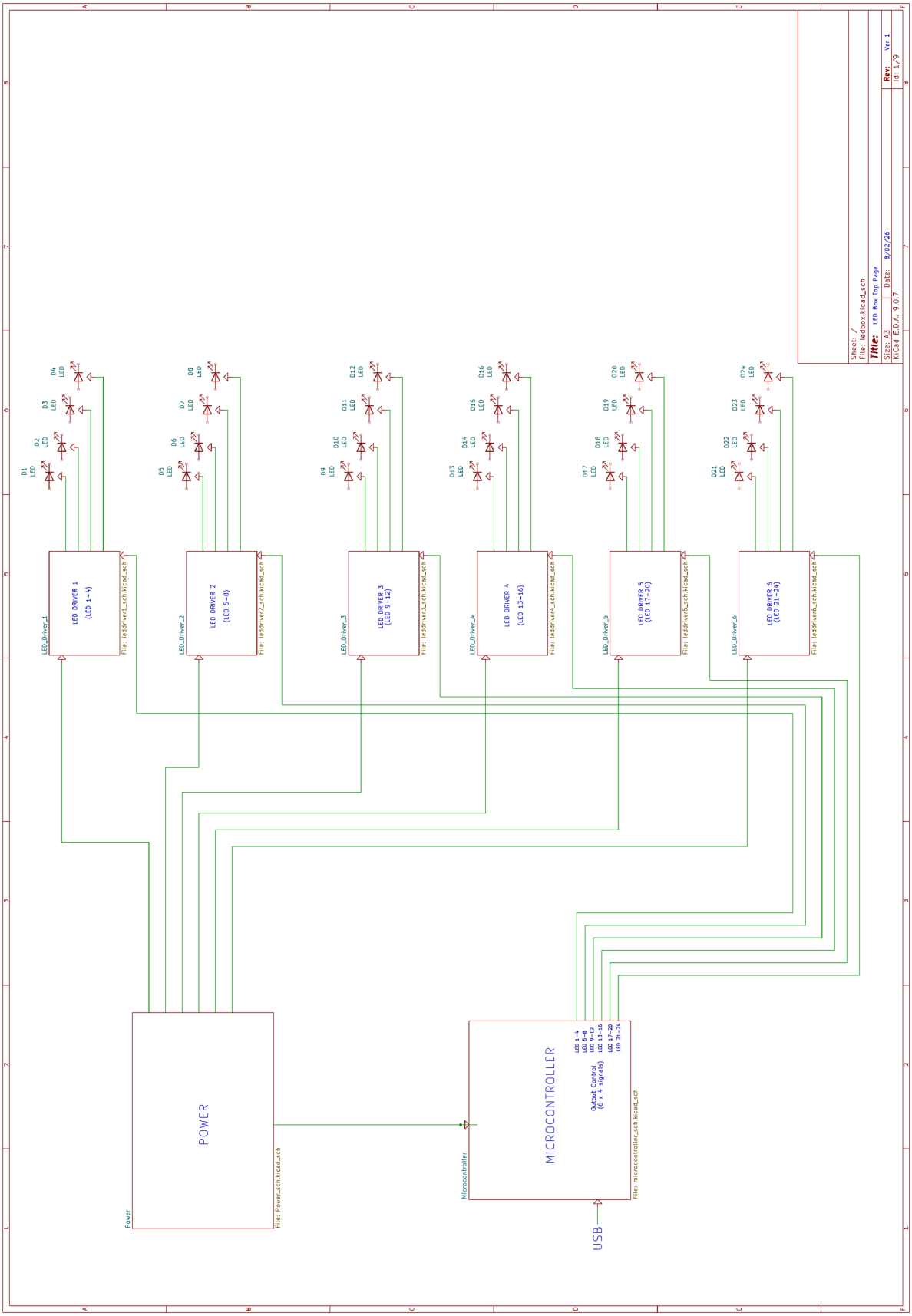
The LED current adjustment range available with the 25 turn trimpot is approximately: **2mA to 17 mA**.

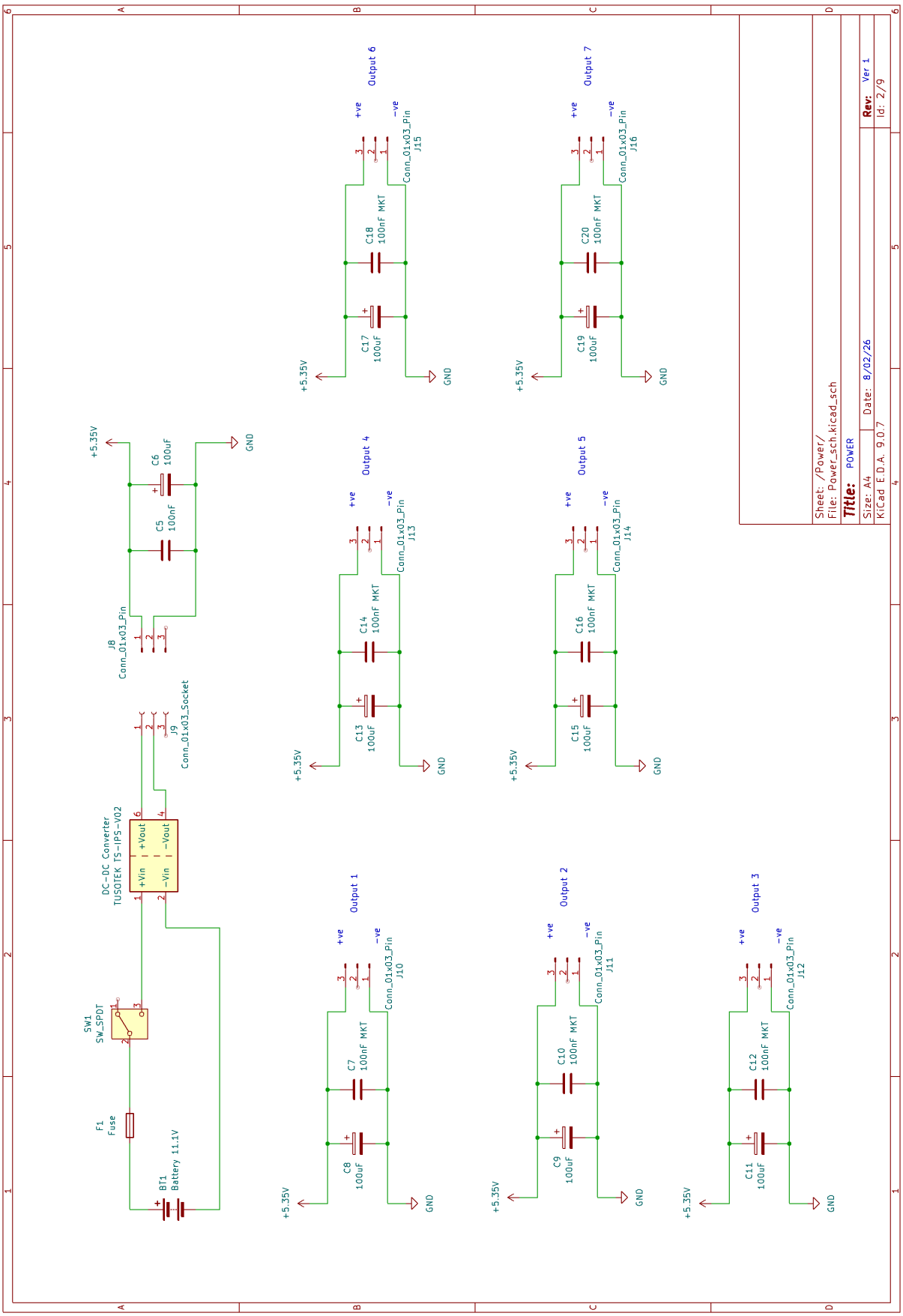
The exact range will depend on the particular 74AHCT125 output high voltage (V_{OH}). Since the LED current is ultimately derived from the available output high voltage of the 74AHCT125, any variation in V_{OH} directly affects the maximum attainable LED current. Lower V_{OH} will reduce the upper current limit, while higher V_{OH} will increase it.

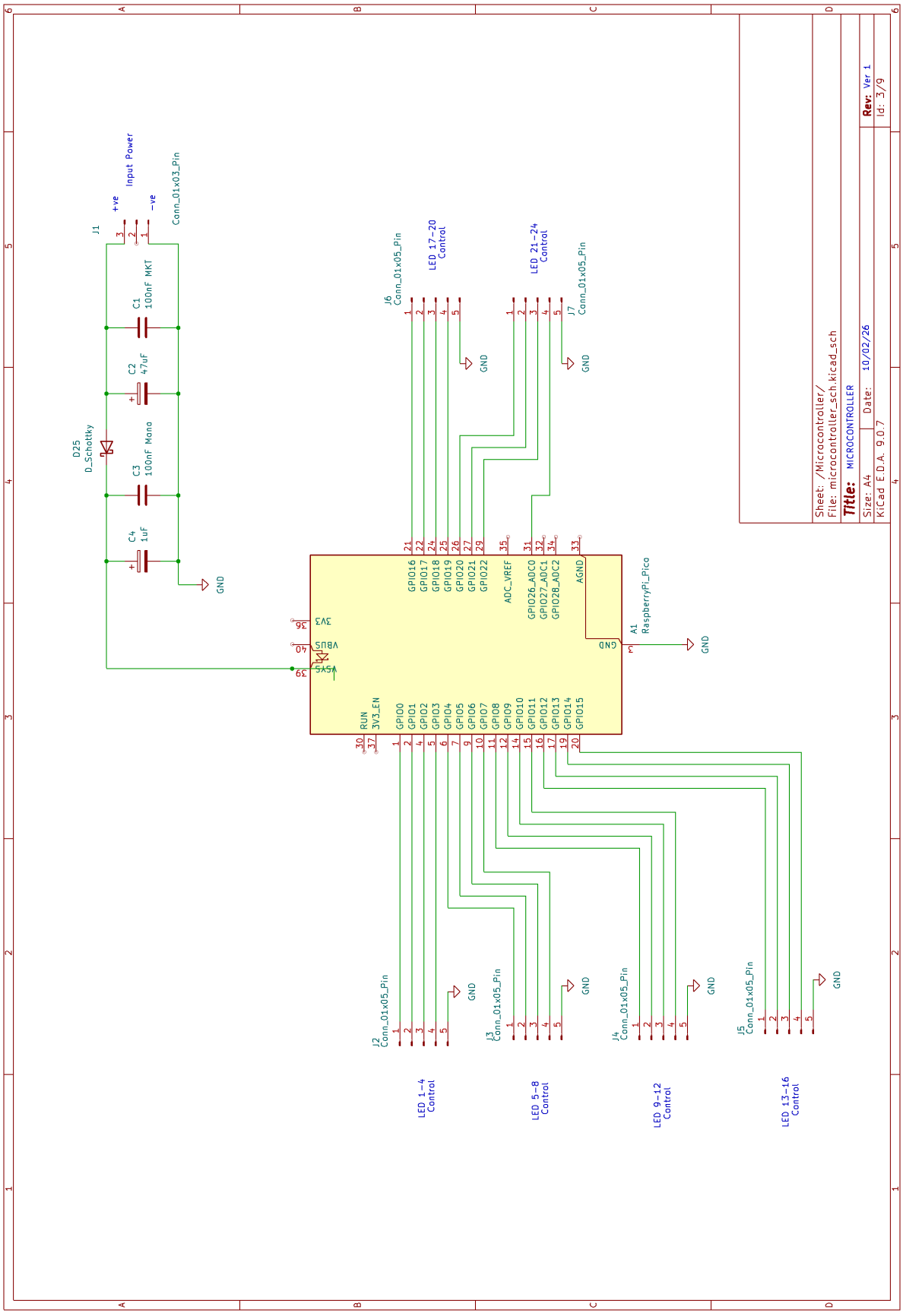
The exact achievable current range is not fixed and will vary due to:

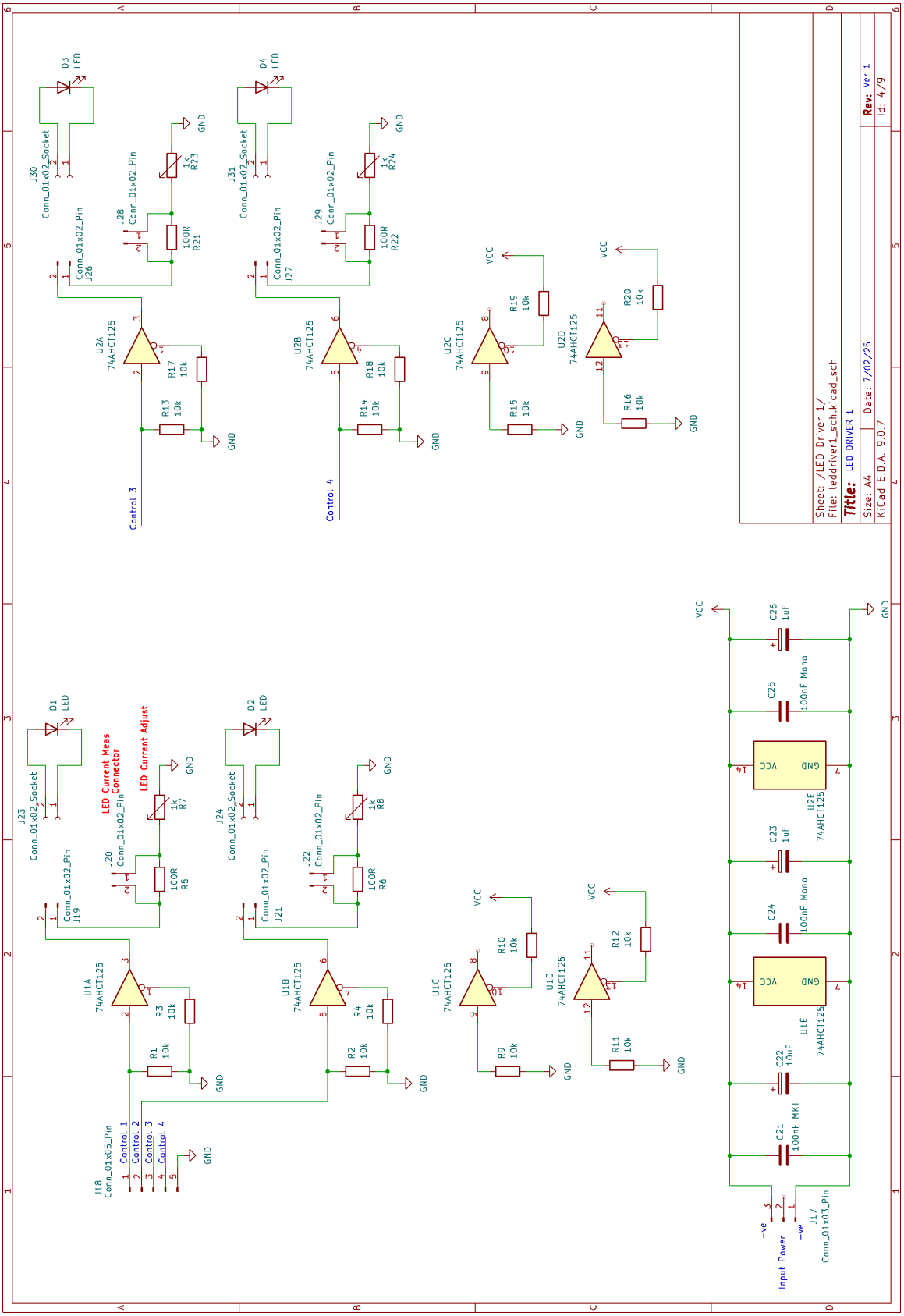
- Device-to-device variation in V_{OH}
- Output loading conditions
- Supply voltage tolerance
- Temperature variation
- Manufacturing spread

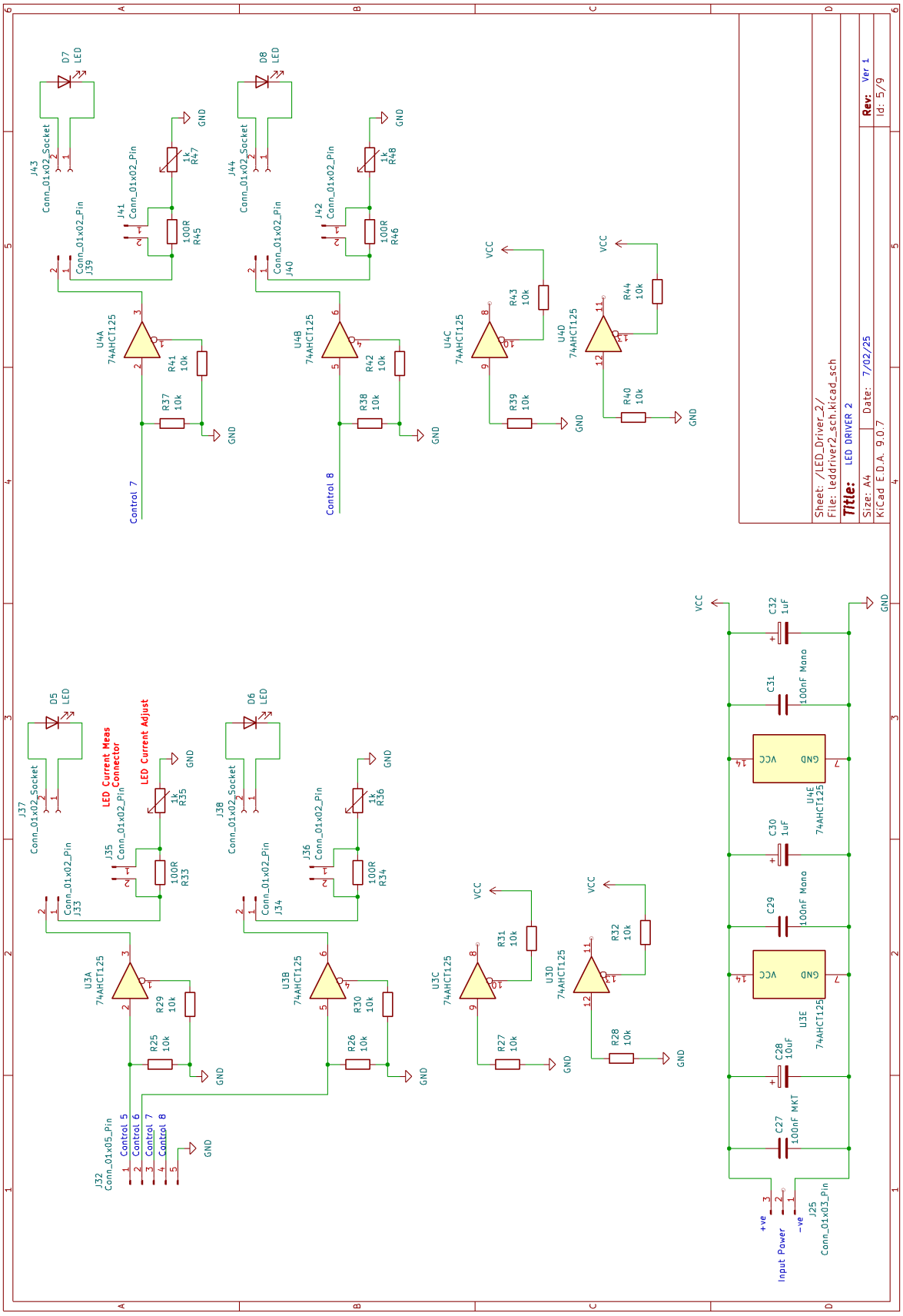
6. Appendix A: Modified LED Box Circuit Diagram

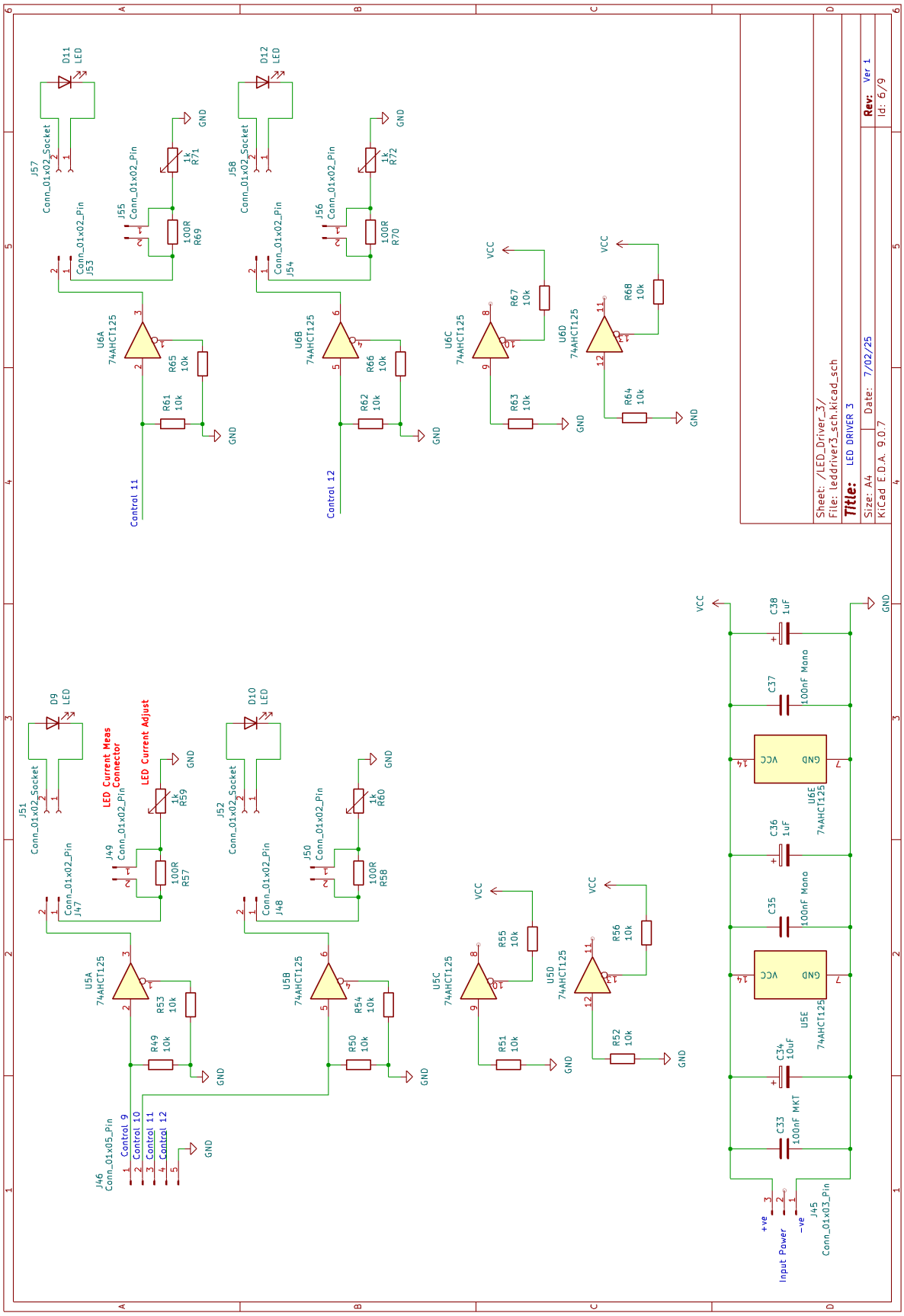

LED DRIVER 1
(LED 1-4)
LED DRIVER 2
(LED 5-8)
LED DRIVER 3
(LED 9-12)
LED DRIVER 4
(LED 13-16)
LED DRIVER 5
(LED 17-20)
LED DRIVER 6
(LED 21-24)
POWER
MICROCONTROLLER
USB
Sheet: /
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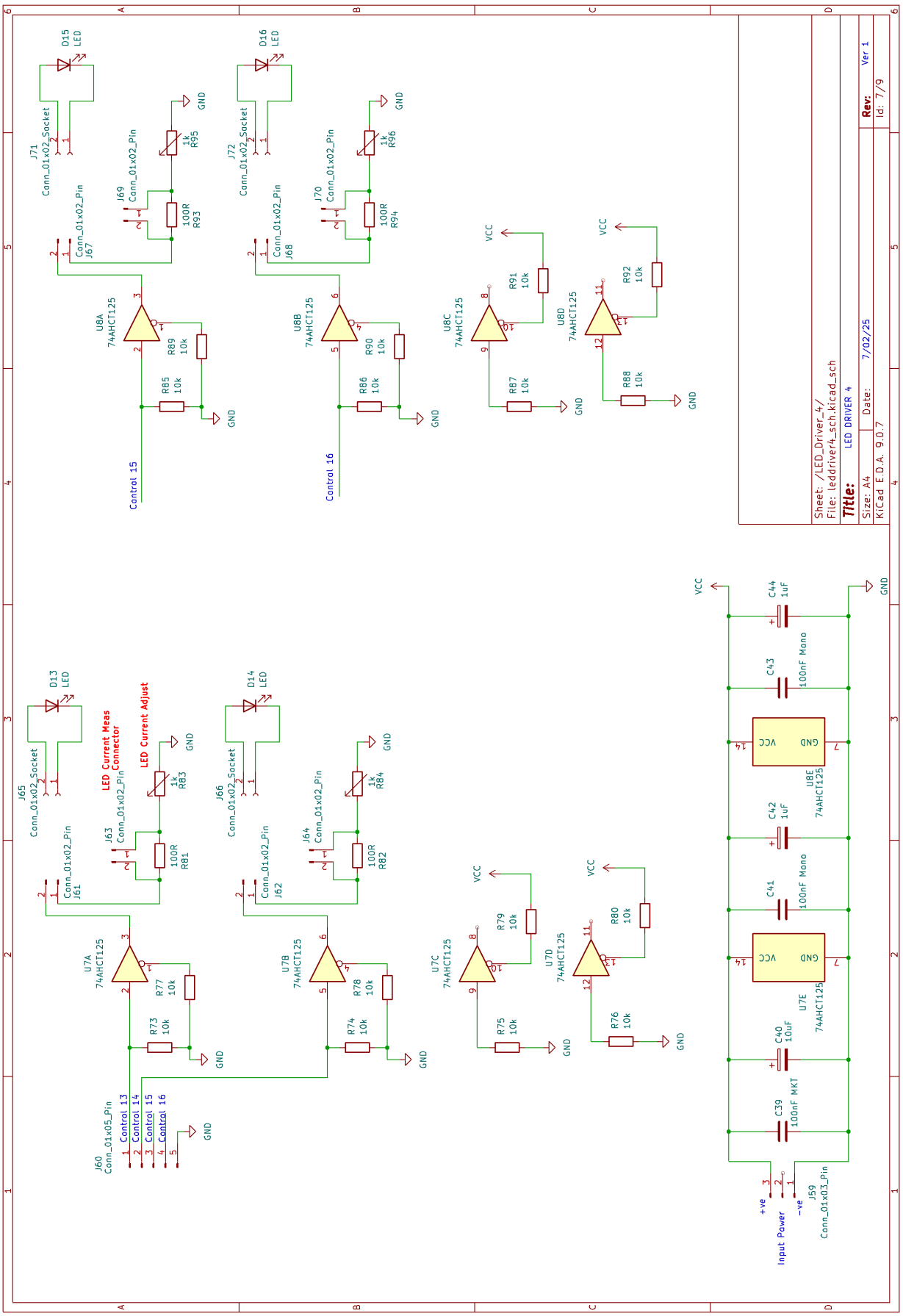

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Output 7
Output 4
Output 5
Output 1
Output 2
Output 3
+ve
-ve
Conn_01x03_Pin
J15
J16
J13
J14
J10
J11
J12
C18 100nF MKT
C20 100nF MKT
C14 100nF MKT
C16 100nF MKT
C7 100nF MKT
C10 100nF MKT
C12 100nF MKT
C17 100uF
C19 100uF
C13 100uF
C15 100uF
C8 100uF
C9 100uF
C11 100uF
+5.35V
GND
J8
Conn_01x03_Pin
C6 100uF
C5 100nF
J9
Conn_01x03_Socket
DC-DC Converter
TUSOTEK TS-1PS-V02
+Vin
-Vin
+Vout
-Vout
SW1
SW_SPDT
F1
Fuse
BT1
Battery 11.1V

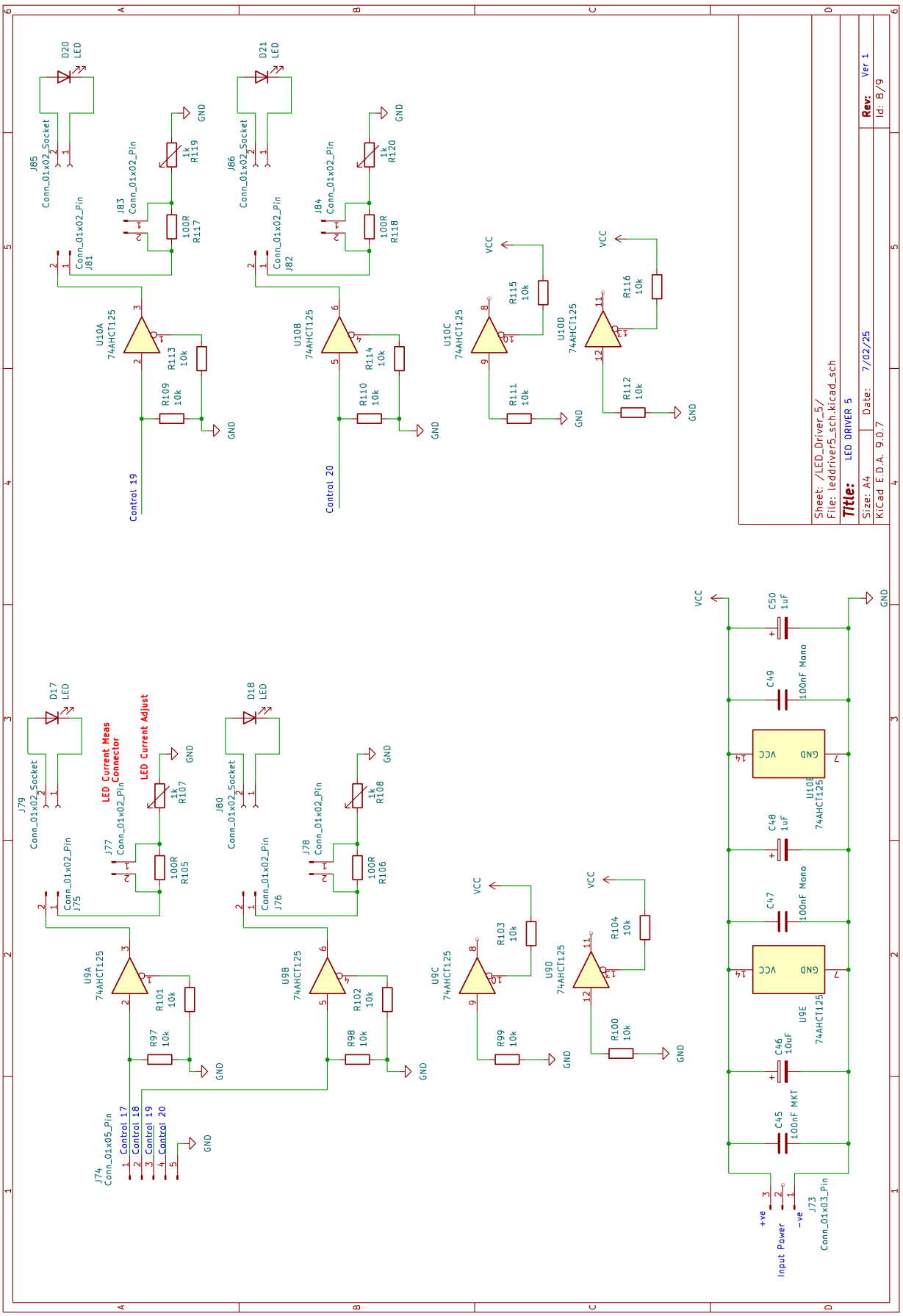

Input Power
+ve
-ve
Conn_01x03_Pin
J1
C1
100nF MKT
C2
47uF
C3
100nF Mono
C4
1uF
D25
D_Schottky
GND
J6
Conn_01x05_Pin
LED 17-20
Control
J7
Conn_01x05_Pin
LED 21-24
Control
GPIO16
GPIO17
GPIO18
GPIO19
GPIO20
GPIO21
GPIO22
ADC_VREF
GPIO26_ADC0
GPIO27_ADC1
GPIO28_ADC2
AGND
GND
A1
RaspberryPi_Pico
3V3
VBUS
VSYS
RUN
3V3_EN
GPIO0
GPIO1
GPIO2
GPIO3
GPIO4
GPIO5
GPIO6
GPIO7
GPIO8
GPIO9
GPIO10
GPIO11
GPIO12
GPIO13
GPIO14
GPIO15
J2
Conn_01x05_Pin
LED 1-4
Control
J3
Conn_01x05_Pin
LED 5-8
Control
J4
Conn_01x05_Pin
LED 9-12
Control
J5
Conn_01x05_Pin
LED 13-16
Control
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Size: A4
Date: 10/02/26
Rev: Ver 1
KiCad E.D.A. 9.0.7
Id: 3/9

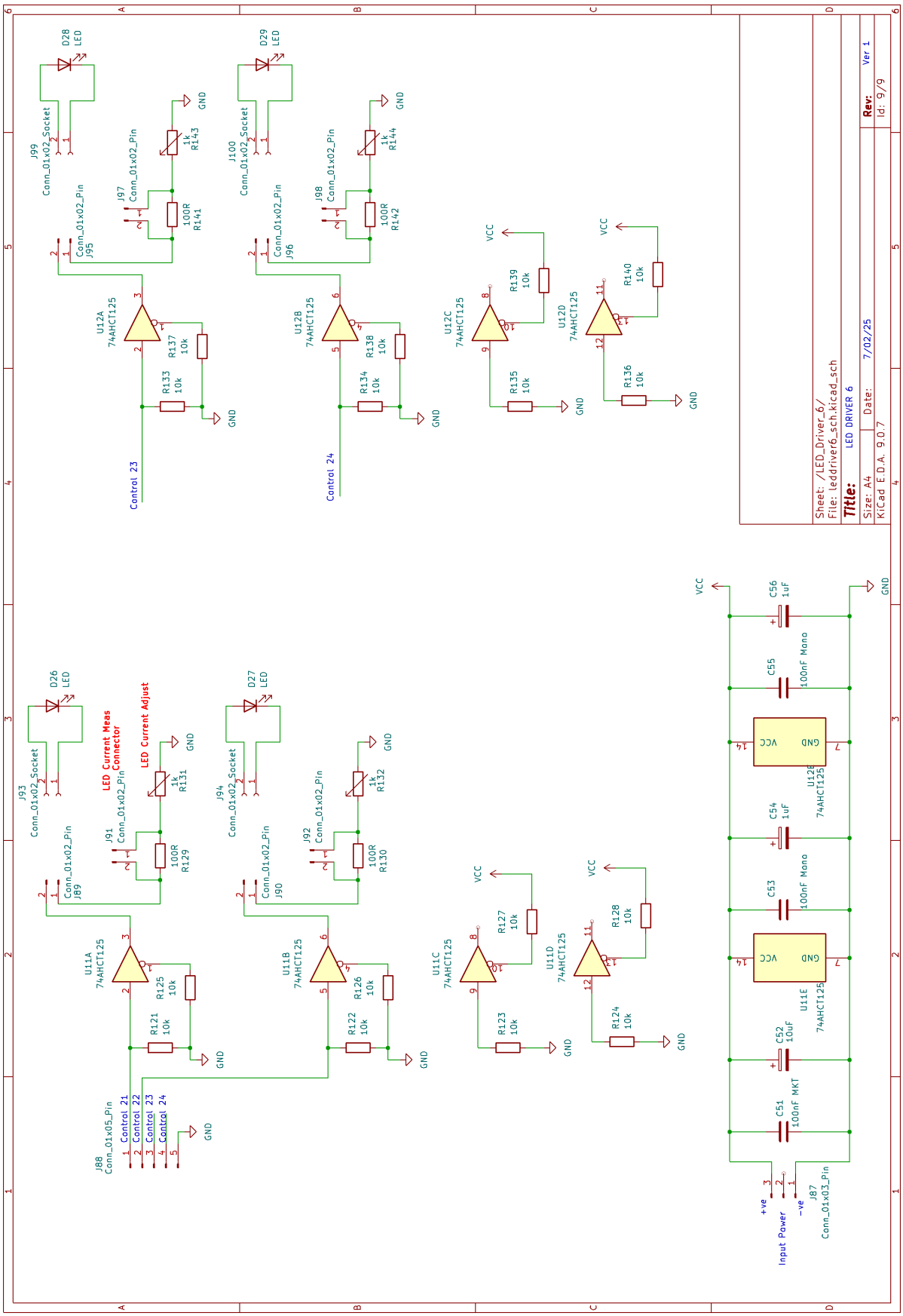

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Rev: Ver 1
KiCad E.D.A. 9.0.7
Id: 4/9
LED Current Meas Connector
LED Current Adjust
Control 3
Control 4
Input Power
+ve
-ve
VCC
GND


Sheet: /LED_Driver_2/
File: leddriver2_sch.kicad_sch
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Size: A4
Date: 7/02/25
Rev: Ver 1
KiCad E.D.A. 9.0.7
Id: 5/9
Control 7
Control 8
LED Current Meas Connector
LED Current Adjust
Input Power
+ve
-ve
GND
VCC


Sheet: /LED_Driver_3/
File: leddriver3_sch.kicad_sch
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Size: A4
Date: 7/02/25
Rev: Ver 1
KiCad E.D.A. 9.0.7
Id: 6/9
Input Power
LED Current Meas Connector
LED Current Adjust
Control 11
Control 12
GND
VCC


Sheet: /LED_Driver_4/
File: leddriver4_sch.kicad_sch
Title: LED DRIVER 4
Size: A4
Date: 7/02/25
KiCad E.D.A. 9.0.7
Rev: Ver 1
Id: 7/9
LED Current Meas Connector
LED Current Adjust
Control 15
Control 16
Input Power
+ve
-ve
GND
VCC


Sheet: /LED_Driver_5/
File: leddriver5_sch.kicad_sch
Title: LED DRIVER 5
Size: A4
Date: 7/02/25
Rev: Ver 1
KiCad E.D.A. 9.0.7
Id: 8/9
LED Current Meas Connector
LED Current Adjust
Control 19
Control 20
Input Power
+ve
-ve
GND
VCC


Sheet: /LED_Driver_6/
File: leddriver6_sch.kicad_sch
Title: LED DRIVER 6
Size: A4
Date: 7/02/25
Rev: Ver 1
KiCad E.D.A. 9.0.7
Id: 9/9
LED Current Meas Connector
LED Current Adjust
Control 23
Control 24
Input Power
+ve
-ve
VCC
GND

7. Appendix B: Component Type Summary

Table B1: Component Type Summary

Component/Part		Supplier and Part Number
Capacitor: 100uF Electrolytic 63V Low ESR		Electus/Jaycar Cat No: RE6346
Capacitor: 100nF MKT		Electus/Jaycar Cat No: RM7125
Capacitor: 100nF Monolithic		Electus/Jaycar Cat No: RC5490
Capacitor: 47uF Electrolytic 63V Low ESR		Electus/Jaycar Cat No: RE6344
Capacitor: 1uF Tantulum 35V		Electus/Jaycar Cat No: RZ6628
Capacitor: 10uF Electrolytic 63V		Electus/Jaycar Cat No: RE6075
Resistor: 10kΩ 1% 0.5W Metal Film		Electus/Jaycar Cat No: RR0596
Resistor: 100Ω 1% 0.5W Metal Film		Electus/Jaycar Cat No: RR0548
Resistor: 1kΩ 25 turn trimpot		Electus/Jaycar Cat No: RT4644
Texas Instruments 74AHCT125		Core Electronics SKU: ADA1787
14 Pin machined IC socket		Electus/Jaycar Cat No: PI6454
1N5819 Schottky Diode, STMicroelectronics		RS Components: 4864460
Raspberry Pi Pico Microcontroller		Core Electronics SKU: CE08702
2 Pin Plug, RS Pro		RS Components: 8971159
2 Pin Socket, RS Pro		RS Components: 8971183
3 Pin Plug, RS Pro		RS Components: 8971168
3 Pin Socket, RS Pro		RS Components: 8971187
5 Pin Plug, RS Pro		RS Components: 8971165
5 Pin Socket, RS Pro		RS Components: 8971199
Male header pin strips		Electus/Jaycar Cat No: HM3212
Female header pins (1x20)		Core Electronics SKU: CE07759
Adafruit 6068 USB Micro B Small Round Panel Mount Extension USB Cable		Core Electronics SKU: ADA6068
Fuseholder: Multicomp Pro CFH05		Element14: 1437178
Fuse: Littlefuse 0215001.HXP		Element14: 1149650
Small proto-board		Electus/Jaycar Cat No: HP9550
Medium proto-board		Electus/Jaycar Cat No: HP9552
Male-Male 40cm Jumper Leads		Core Electronics SKU: CE09614
Male-Male 60cm Jumper Leads		Robot Gear SKU: 010008
Existing Components		
DC-DC Converter		Unknown
On/off switch DPDT		Unknown
LEDs		Unknown

8. Appendix C: LED Current Setting Uncertainty Analysis

To meet the design requirement of driving the LEDs at the same current level as the existing LED Box (7–8 mA), all LEDs have been set to a nominal current of 7.5 mA. This corresponds to a measured voltage of 0.750 V across a 100Ω ±1% precision resistor.

The voltage drop across the 100Ω resistor was measured using a Rigol DM3058E 5½-digit bench multimeter on the 2.00000 V range. The manufacturer's specified accuracy [3] (see data sheet extract below) for this range is:

± (0.015% of reading + 0.003% of range)

For a measured value of 0.750 V:

- 0.015% of 0.750 V = 0.0001125 V
- 0.003% of 2.00000 V = 0.00006 V

    ⇨ Total worst case error ≈ ±0.0001725 V (≈ ±0.173 mV)

This corresponds to a current uncertainty of:

$$\frac{0.173\ \text{mV}}{100\Omega} \approx \pm 0.0017\ \text{mA}$$

This error contribution is negligible compared to the resistor tolerance of ±1%, which alone introduces a potential current variation of:

$$\pm 1\% \times 7.5\ \text{mA} = \pm 0.075\ \text{mA}$$

Therefore, the dominant source of uncertainty in the LED current setting is the resistor tolerance, not the multimeter accuracy.

In practice, also considering the adjustment granularity of the 25-turn trimpot, the achievable current setting is approximately:

$$7.5 \pm 0.1\ \text{mA}$$

This satisfies the design requirement of setting the LEDs operating current within the 7–8 mA range.

Specifications

DC Characteristics

Accuracy ± (% of reading + % of range)[1]

Function	Range[2]	Test current or Load voltage	1 Year 23℃± 5℃	Temperature Coefficient 0℃ - 18℃ 28℃ - 50℃
DC Voltage	200.000 mV		0.015 + 0.004	0.0015 + 0.0005
	2.00000 V		0.015 + 0.003	0.0010 + 0.0005
	20.0000 V		0.015 + 0.004	0.0020 + 0.0005
	200.000 V		0.015 + 0.003	0.0015 + 0.0005
	1000.00 V[4]		0.015 + 0.003	0.0015 + 0.0005
DC Current	200.000 µA	<8 mV	0.055 + 0.005	0.003 + 0.001
	2.00000 mA	<80 mV	0.055 + 0.005	0.002 + 0.001
	20.0000 mA	<0.05 V	0.095 + 0.020	0.008 + 0.001
	200.000 mA	<0.5 V	0.070 + 0.008	0.005 + 0.001
	2.00000 A	<0.1 V	0.170 + 0.020	0.013 + 0.001
	10.0000 A[5]	<0.3 V	0.250 + 0.010	0.008 + 0.001
Resistance[3]	200.000 Ω	1 mA	0.030 + 0.005	0.0030 + 0.0006
	2.00000 kΩ	1 mA	0.020 + 0.003	0.0030 + 0.0005
	20.0000 kΩ	100 µA	0.020 + 0.003	0.0030 + 0.0005
	200.000 kΩ	10 µA	0.020 + 0.003	0.0030 + 0.0005
	2.00000 MΩ	1 µA	0.040 + 0.004	0.0040 + 0.0005
	10.0000 MΩ	200 nA	0.250 + 0.003	0.0100 + 0.0005
	100.000 MΩ	200 nA || 10 MΩ	1.75 + 0.004	0.2000 + 0.0005
Diode Test	2.0000 V[6]	1 mA	0.05 + 0.01	0.0050 + 0.0005
Continuity Test	2000 Ω	1 mA	0.05 + 0.01	0.0050 + 0.0005

Remarks:

[1] Specifications are for 0.5 hour warm-up, "Slow" measurement rate and calibration temperature 18℃ ~ 28℃.

[2] 20% over range on all ranges except for DCV 1000 V, ACV 750 V, DCI 10 A and ACI 10 A.

[3] Specifications are for 4-wire measure or 2-wire measure under "REF" operation. ± 0.2 Ω of extra errors will be generated if perform 2-wire measure without "REF" operation.

[4] Plus 0.02 mV of error per 1 V after the first ±500 VDC.

[5] 30 seconds OFF after 30 seconds ON is recommend for the continuous current that higher than DC 7 A or AC RMS 7 A.

[6] Accuracy specifications are only for voltage measuring at input terminal. The typical value of current under measure is 1 mA. Voltage drop at the diode junction may vary with current supply.

Figure C1: Rigol DM3058E Digital Multimeter DC Accuracy Specifications (extracted from [3], the Rigol DM3045E Data Sheet)

9. Document History

Date	Version	Author	Comments
28 February 2026	Version 1	E.Scipione/X. Zheng	Initial Release of Version 1